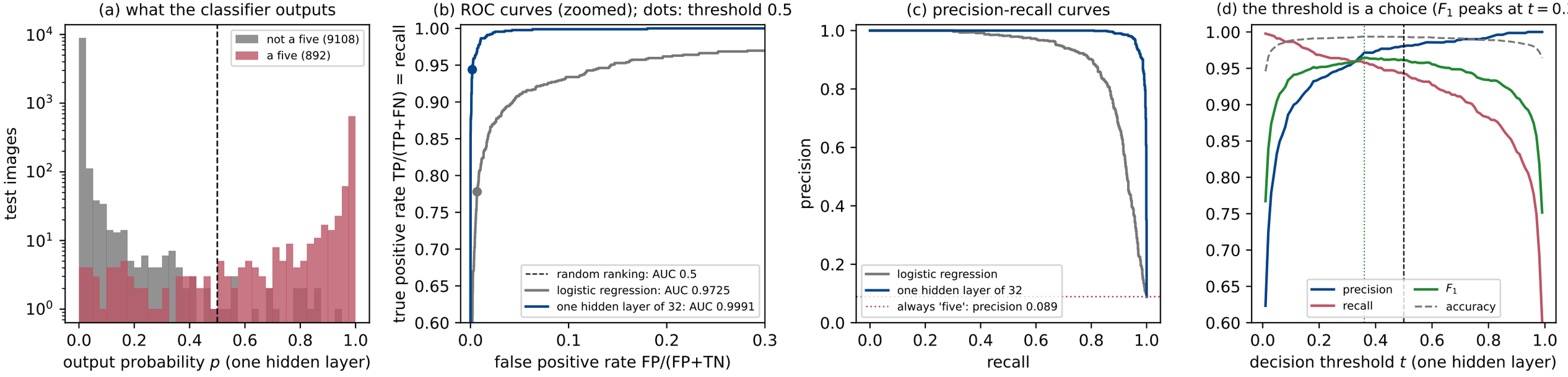
(a) what the classifier outputs
not a five (9108)
a five (892)
test images
output probability p (one hidden layer)
(b) ROC curves (zoomed); dots: threshold 0.5
true positive rate TP/(TP+FN) = recall
false positive rate FP/(FP+TN)
random ranking: AUC 0.5
logistic regression: AUC 0.9725
one hidden layer of 32: AUC 0.9991
(c) precision-recall curves
precision
recall
logistic regression
one hidden layer of 32
always 'five': precision 0.089
(d) the threshold is a choice (F_1 peaks at $t = 0.36$)
decision threshold t (one hidden layer)
precision
recall
F_1
accuracy